Linear optimization problems

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Outline

Linear Programming problems

Examples and properties of linear optimization problems

Linear Programming problems

Introduction: Linear Programming problems

- In this section we define what a **Linear Programming (LP) problem** is: optimize a linear function over a polyhedron
- We give a name to every ingredient: feasible region, objective function, input data, optimal solutions and optimal value
- We see when an optimal solution does not exist, and how a feasible solution gives a bound on the optimal value
- These are the words we will use in the whole course

Definition 1: Linear Programming problem in minimization and maximization form

*Given a polyhedron $F \subseteq \mathbb{R}^n$ and a linear function $f : \mathbb{R}^n \to \mathbb{R}$, the associated **Linear Programming problem** LP consists of,*

- *in the **minimization form**, finding a point $\tilde{\boldsymbol{x}}(LP) \in F$ such that*

$$f(\tilde{\boldsymbol{x}}(LP)) \leq f(\boldsymbol{x}), \qquad \forall \boldsymbol{x} \in F;$$

- *in the **maximization form**, finding a point $\tilde{\boldsymbol{x}}(LP) \in F$ such that*

$$f(\tilde{\boldsymbol{x}}(LP)) \geq f(\boldsymbol{x}), \qquad \forall \boldsymbol{x} \in F.$$

An LP problem has two ingredients:

a **polyhedron** F and a **linear function** f.

Let us give them a name, and see which data describe them.

Feasible region and objective function

- The polyhedron F is called the **feasible region** of the LP problem
- Its points $\boldsymbol{x} \in F$ are called the **feasible solutions** of the LP problem
- The linear function f is the **objective function** of the LP problem
- It assigns an objective value to each feasible solution

The data of the feasible region

- To define the polyhedron of the feasible region of an LP problem, we need the number m of its inequalities/equalities and
  1. a matrix $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ — the **constraint matrix**,
  2. a (column) vector $\boldsymbol{b} \in \mathbb{R}^m$ — the **right-hand-side (RHS) vector**.

$$\underbrace{\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}}_{\boldsymbol{A}} \in \mathbb{R}^{m \times n} \qquad \underbrace{\begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}}_{\boldsymbol{b}} \in \mathbb{R}^m$$

- Moreover, for each row we need to specify whether it corresponds to a half-space (linear inequality) or a hyperplane (linear equality)

The data of the objective function

- To define the linear objective function of an LP problem, we need the (column) vector $\boldsymbol{c} \in \mathbb{R}^n$, called the **objective-function vector**:

$$\underbrace{\begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}}_{\boldsymbol{c}} \in \mathbb{R}^n.$$

The input data of an LP problem

The **input data** of an LP problem are:

- the constraint matrix $\boldsymbol{A} \in \mathbb{R}^{m \times n}$
- the right-hand-side vector $\boldsymbol{b} \in \mathbb{R}^{m}$
- the objective-function vector $\boldsymbol{c} \in \mathbb{R}^{n}$
- for each row, the information specifying whether the corresponding constraint is an equality or an inequality

Now that we know the ingredients,
we name the solutions we are looking for
and their objective value.

Definition 2: optimal solution and optimal value of an LP problem (1/2)

- *A point $\tilde{\boldsymbol{x}}(LP) \in F$ as in the definition of the LP problem is called an **optimal solution** of the LP problem.*
  1. *In the minimization form,*

     $$\tilde{\boldsymbol{x}}(LP) \in \arg\min\{\, f(\boldsymbol{x}) \text{ subject to } \boldsymbol{x} \in F \,\}.$$

  2. *In the maximization form,*

     $$\tilde{\boldsymbol{x}}(LP) \in \arg\max\{\, f(\boldsymbol{x}) \text{ subject to } \boldsymbol{x} \in F \,\}.$$

Definition: optimal solution and optimal value of an LP problem (2/2)

- *The value $f(\tilde{\boldsymbol{x}}(LP))$ is called the **optimal value** of the LP problem, and it is denoted by $z(LP)$.*
  1. *In the minimization form,*

$$z(LP) = \underbrace{\min\{\, f(\boldsymbol{x}) \text{ subject to } \boldsymbol{x} \in F \,\}}_{=f(\tilde{\boldsymbol{x}}(LP))}.$$

  2. *In the maximization form,*

$$z(LP) = \underbrace{\max\{\, f(\boldsymbol{x}) \text{ subject to } \boldsymbol{x} \in F \,\}}_{=f(\tilde{\boldsymbol{x}}(LP))}.$$

Many optimal solutions, or none

- An LP problem may admit multiple (possibly infinitely many) optimal solutions, or none at all
- In such cases, the operators arg min and arg max return the full set of optimal solutions when they exist, and return the empty set otherwise

With a slight abuse of notation:

1. In the **minimization form**, we write the LP problem as

$$\min\{\, f(\boldsymbol{x}) \text{ subject to } \boldsymbol{x} \in F \,\}.$$

2. In the **maximization form**, we write the LP problem as

$$\max\{\, f(\boldsymbol{x}) \text{ subject to } \boldsymbol{x} \in F \,\}.$$

The goal of an LP problem

Although we use the symbols "min" and "max",
the goal of an LP problem is to **find an optimal solution** $\tilde{x}(LP)$,
whenever it exists,

rather than merely determining the optimal value $z(LP)$.

Points of minimum and points of maximum

- In the **minimization form**:
  - an optimal solution is a **point of minimum**
  - the corresponding optimal value is the **minimum** of the objective function over the feasible region
- Similarly, in the **maximization form**:
  - an optimal solution is a **point of maximum**
  - the corresponding optimal value is the **maximum** of the objective function over the feasible region

LP problems without an optimal solution

An optimal solution of an LP problem may not exist for one of the following two reasons:

1. If the objective function decreases indefinitely in the minimization form or increases indefinitely in the maximization form, the LP problem is said to be **unbounded**

   In these cases, we write $z(LP) = -\infty$ and $z(LP) = +\infty$, respectively

2. If the feasible region is empty, meaning that no feasible solutions exist, the LP problem is said to be **infeasible**

   It has no optimal solution and $z(LP)$ is not defined

There is no third reason (1/2)

- If an LP problem has feasible solutions and it is not unbounded, the infimum (in the minimization form) or the supremum (in the maximization form) of the objective function over the feasible region is always **attained** by a feasible solution

- This is due to the special structure of LP problems:
  - the objective function is linear
  - the feasible region is a polyhedron, a **closed** set defined by linear inequalities "$\leq$", "$\geq$" and linear equalities "$=$"
  - never by strict inequalities "$<$", "$>$"

There is no third reason (2/2)

- For example, the interval

$$(0, 1] = \{ x \in \mathbb{R} : 0 < x \leq 1 \}$$

  is not a polyhedron, and

$$\min\{ x \text{ subject to } 0 < x \leq 1 \}$$

  is not an LP problem: its infimum 0 is not attained

- The three possible outcomes of an LP problem (infeasible, unbounded, optimal solution) are formally defined and studied in the chapter «Possible outcomes and equivalent formulations»

Example (1/2)

LP problems without an optimal solution.

Consider the objective function $f(x) = 2x + 1$ and the feasible region $F = \{ x \in \mathbb{R} : x \geq -2 \}$, a half-line.

- **Minimization form:** $2x + 1 \geq -3$ for every feasible x, with equality only for $x = -2$
  - the unique optimal solution is $\tilde{\boldsymbol{x}}(LP) = (-2)$ and $z(LP) = -3$
- **Maximization form:** for every value $M \geq 0$ the feasible solution $x = M$ has

$$f(M) = 2M + 1 > M$$

  - the objective function increases indefinitely and the problem is **unbounded**, $z(LP) = +\infty$

Example (2/2)

- Consider the feasible region

$$F = \{ x \in \mathbb{R} : x \leq -1,\ x \geq 1 \}$$

- No real number satisfies both inequalities, so $F = \emptyset$
- For every linear objective function, both the minimization and the maximization form are **infeasible**

Even before finding an optimal solution,
every feasible solution already tells us something
about the optimal value.

Feasible solutions and bounds on the optimal value

- Given a **feasible solution** of an LP problem

$$\underbrace{\begin{pmatrix} \bar{x}_1 \\ \bar{x}_2 \\ \vdots \\ \bar{x}_n \end{pmatrix}}_{\bar{\boldsymbol{x}}(LP)} \in F,$$

we have

1. in **minimization** form, the following **upper bound**:

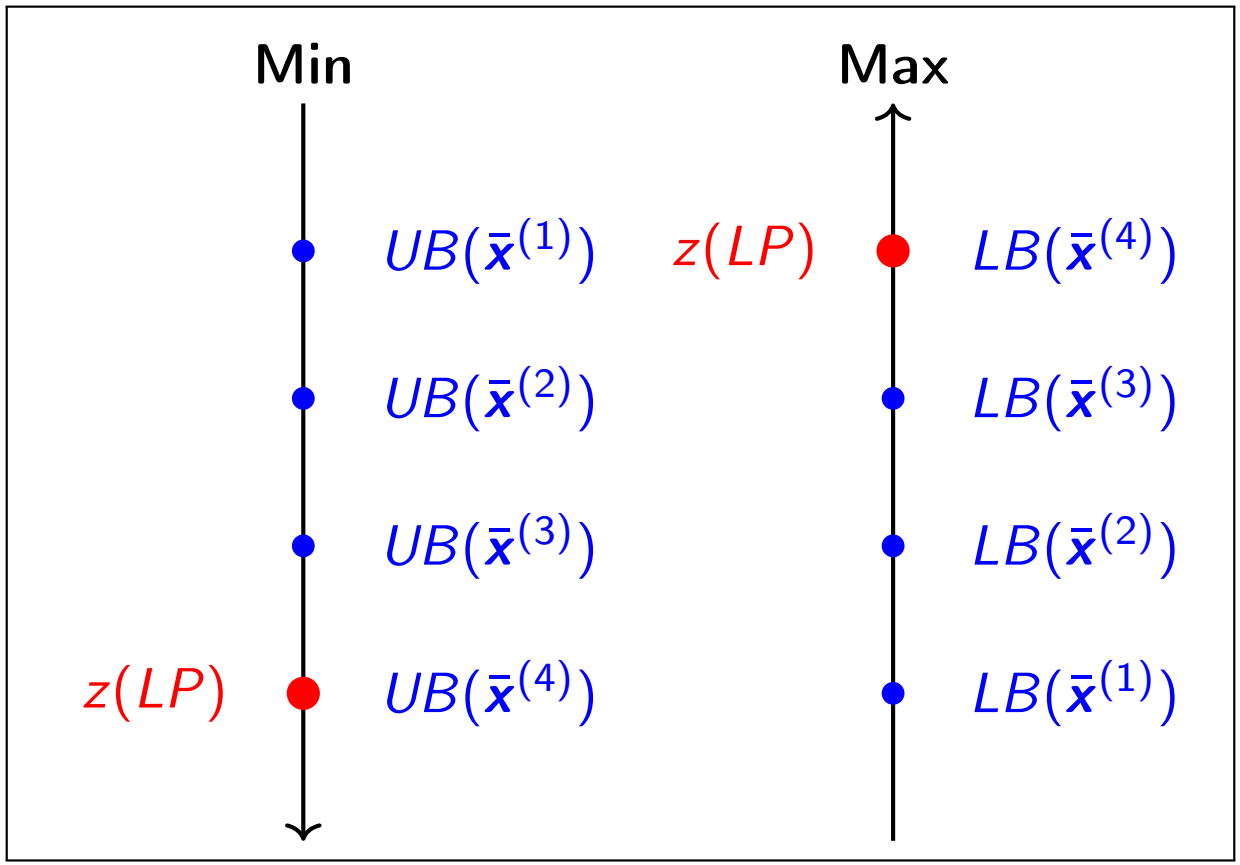
$$z(LP) \leq \underbrace{f(\bar{\boldsymbol{x}}(LP))}_{UB(\bar{\boldsymbol{x}})};$$

2. in **maximization** form, the following **lower bound**:

$$\underbrace{f(\bar{\boldsymbol{x}}(LP))}_{LB(\bar{\boldsymbol{x}})} \leq z(LP).$$

Bounds from four feasible solutions

- Four feasible solutions $\bar{\boldsymbol{x}}^{(1)}, \bar{\boldsymbol{x}}^{(2)}, \bar{\boldsymbol{x}}^{(3)}, \bar{\boldsymbol{x}}^{(4)} \in F$ give lower bounds in the maximization case and upper bounds in the minimization case for $z(LP)$ (the fourth one attains the optimal value)



Summary (1/2)

LP problem:

- Optimize (minimize or maximize) a linear function f over a polyhedron F
- F is the feasible region, its points are the feasible solutions, f is the objective function

Input data:

- The constraint matrix $\boldsymbol{A}$, the RHS vector $\boldsymbol{b}$, the objective-function vector $\boldsymbol{c}$, and the type ($=$ or inequality) of each row

Optimal solutions and optimal value:

- $\tilde{\boldsymbol{x}}(LP)$ is a point of minimum (maximum), $z(LP)$ is its value
- The goal is to find an optimal solution, not only $z(LP)$

Summary (2/2)

No optimal solution:

- Unbounded problem: $z(LP) = -\infty$ (min) or $z(LP) = +\infty$ (max)
- Infeasible problem: $F = \emptyset$, $z(LP)$ is not defined
- No third reason: the feasible region is a closed set, so the infimum (supremum) is attained

Bounds:

- Every feasible solution $\bar{\boldsymbol{x}}$ gives an upper bound $UB(\bar{\boldsymbol{x}})$ in minimization form and a lower bound $LB(\bar{\boldsymbol{x}})$ in maximization form

Examples and properties of linear optimization problems

Introduction: Examples and properties of linear optimization problems

- We first solve a small LP problem in **one variable**, where we can see everything on a graph
- Then we state four **properties** of LP problems: how the optimal value changes when we change the sign of the objective function, add a constant, or multiply it by a constant
- These properties let us move freely between minimization and maximization, and simplify the objective function without changing the optimal solutions

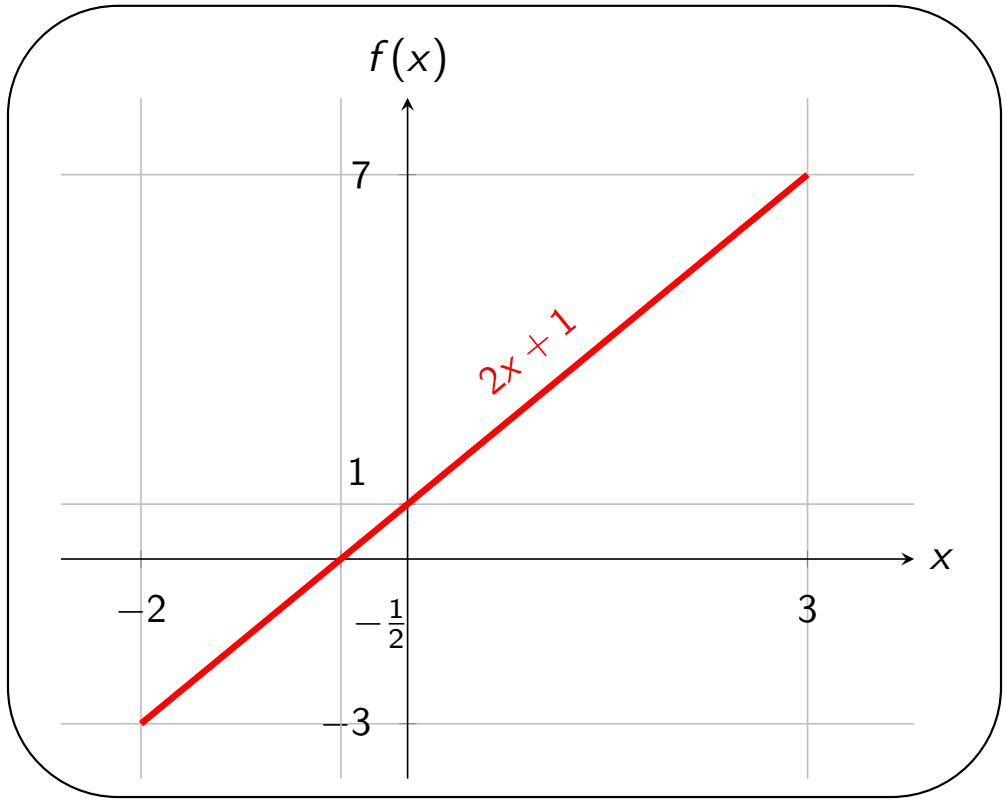
The following example is a linear optimization problem in one variable:

the feasible region is an interval (a polyhedron in $\mathbb{R}$)
and the objective function is linear.

Example (1/3)

LP problem in one variable. Let us consider the following objective function and feasible region:

$$f : \mathbb{R} \to \mathbb{R}, \; f : x \mapsto 2x + 1 \qquad \text{and} \qquad F = [-2, 3]$$



Example (2/3)

The associated LP problems in minimization and maximization forms are:

$$\min \left\{ 2x + 1 \;\text{ subject to }\; -2 \leq x \leq 3 \right\}$$

$$\max \left\{ 2x + 1 \;\text{ subject to }\; -2 \leq x \leq 3 \right\}$$

- For the minimization form, since the linear function is strictly increasing, the unique optimal solution and the optimal value are:

$$\tilde{\boldsymbol{x}}(LP) = (-2) \quad \text{and} \quad z(LP) = -3.$$

- For the maximization form, since the linear function is strictly increasing, the unique optimal solution and the optimal value are:

$$\tilde{\boldsymbol{x}}(LP) = (3) \quad \text{and} \quad z(LP) = 7.$$

Example (3/3)

- The derivative function is constant:

$$f' : \mathbb{R} \to \mathbb{R}, \;\; f' : x \mapsto 2$$

$$f'(x) = 2 > 0 \quad \forall x \in \mathbb{R},$$

- The function is strictly increasing on the whole interval $F = [-2, 3]$: the minimum is attained at the left endpoint $x = -2$, the maximum at the right endpoint $x = 3$

Properties of LP problems

- The following properties hold for every LP problem with feasible region F and objective function $f(\boldsymbol{x}) = \boldsymbol{c}'\boldsymbol{x}$
- We state them using min and max for LP problems that admit an optimal solution
- If one of the two LP problems in an equation is infeasible (unbounded), so is the other one

Property: minimum and maximum

- For every LP problem, we have:

$$\min \left\{ \boldsymbol{c}'\boldsymbol{x} \text{ subject to } \boldsymbol{x} \in F \right\}$$

$$= - \max \left\{ -\boldsymbol{c}'\boldsymbol{x} \text{ subject to } \boldsymbol{x} \in F \right\}$$

- The optimal solutions do not change: a point minimizes $\boldsymbol{c}'\boldsymbol{x}$ over F if and only if it maximizes $-\boldsymbol{c}'\boldsymbol{x}$ over F

Example (1/2)

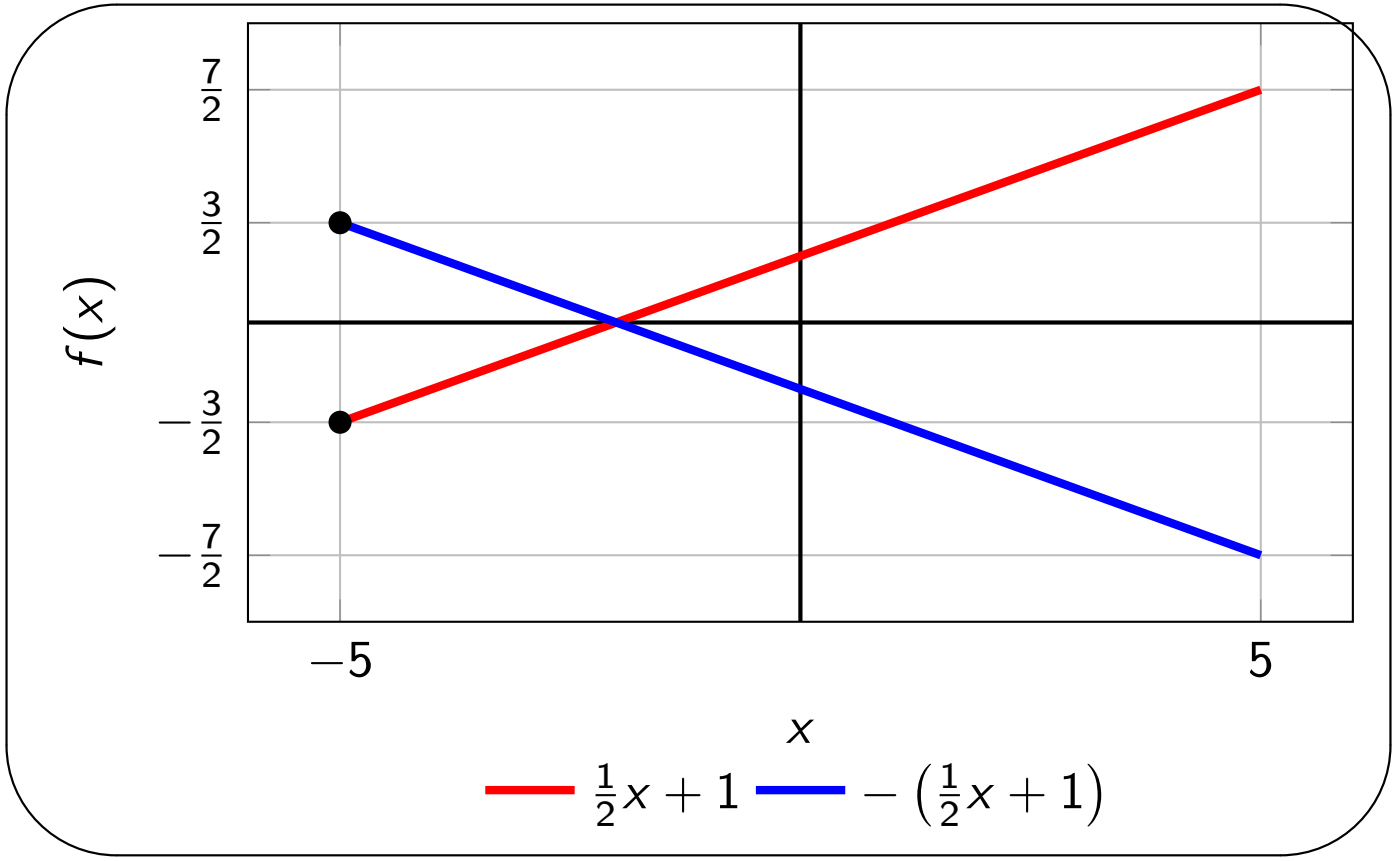
Property: minimum and maximum.

$$\underbrace{\min\left\{\frac{1}{2}x+1 \text{ subject to } -5\leq x\leq 5\right\}}_{=-\frac{3}{2}}$$

$$=-\underbrace{\max\left\{-\left(\frac{1}{2}x+1\right) \text{ subject to } -5\leq x\leq 5\right\}}_{=\frac{3}{2}}$$

- Both optima are attained at $x=-5$ (black points in the figure on the next slide)

Example (2/2)


$\frac{7}{2}$
$\frac{3}{2}$
$-\frac{3}{2}$
$-\frac{7}{2}$
$f(x)$
-5
5
x
$\frac{1}{2}x+1$
$-\left(\frac{1}{2}x+1\right)$

Property: constant addition

- For every LP problem and every $a \in \mathbb{R}$, we have:

$$\min \left\{ a + \boldsymbol{c}'\boldsymbol{x} \text{ subject to } \boldsymbol{x} \in F \right\}$$

$$= a + \min \left\{ \boldsymbol{c}'\boldsymbol{x} \text{ subject to } \boldsymbol{x} \in F \right\}$$

- Adding a constant moves the graph of the objective function up or down: the optimal solutions do not change

Example (1/2)

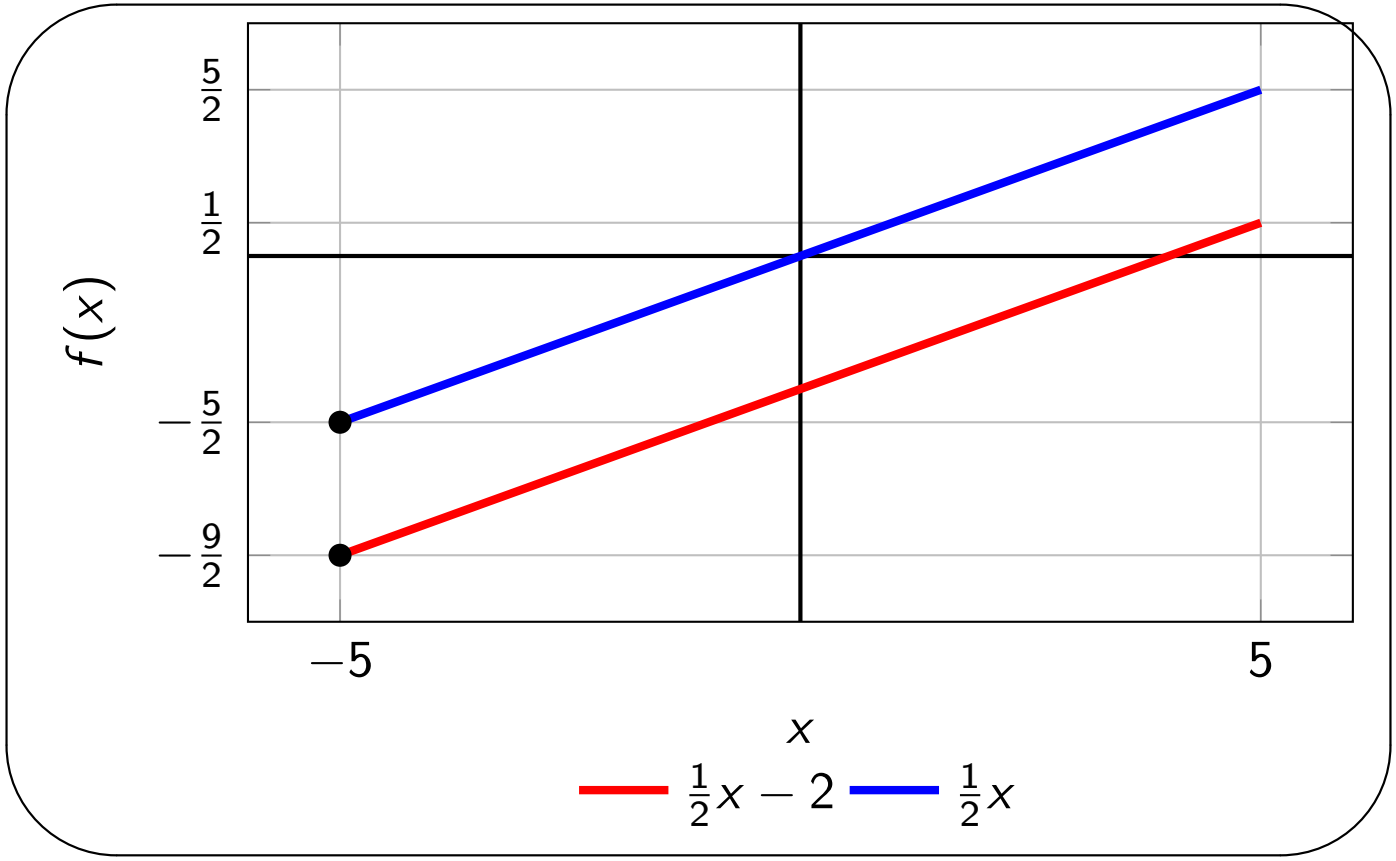
Property: constant addition.

$$\underbrace{\min\left\{\frac{1}{2}x - 2 \text{ subject to } -5 \leq x \leq 5\right\}}_{=-\frac{9}{2}}$$

$$= -2 + \underbrace{\min\left\{\frac{1}{2}x \text{ subject to } -5 \leq x \leq 5\right\}}_{=-\frac{5}{2}}$$

- Both minima are attained at $x = -5$ (black points in the figure on the next slide)

Example (2/2)



Property: multiplication by a positive constant

- For every LP problem and every $a \in \mathbb{R}_{>0}$, we have:

$$\min \left\{ a\, \boldsymbol{c}'\boldsymbol{x} \text{ subject to } \boldsymbol{x} \in F \right\}$$

$$= a \ \min \left\{ \boldsymbol{c}'\boldsymbol{x} \text{ subject to } \boldsymbol{x} \in F \right\}$$

- Multiplying by a positive constant stretches (or compresses) the graph vertically: the optimal solutions do not change

Example (1/2)

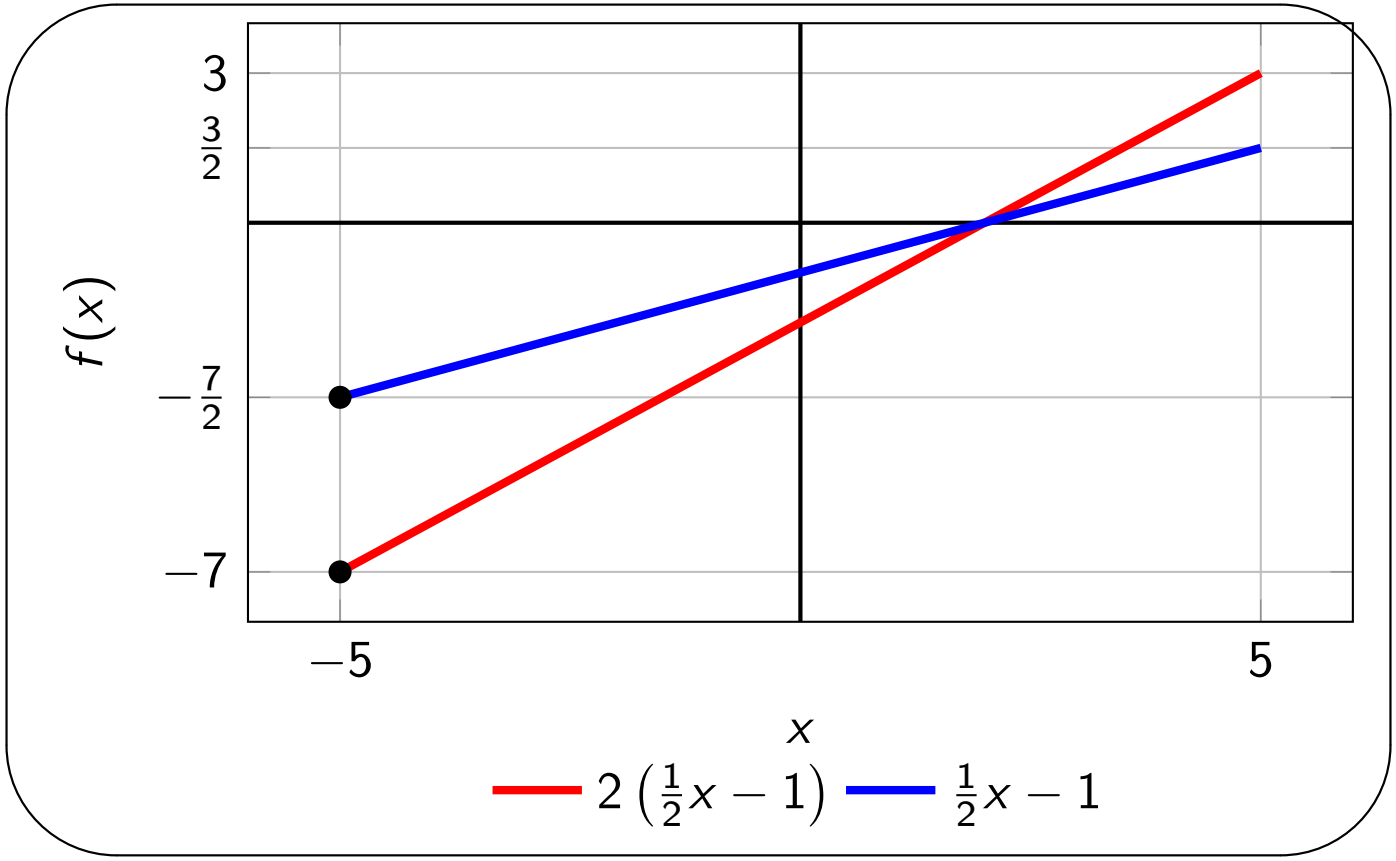
Property: multiplication by a positive constant.

$$\underbrace{\min\left\{ 2\left(\frac{1}{2}x - 1\right) \text{ subject to } -5 \leq x \leq 5 \right\}}_{=-7}$$

$$= 2 \underbrace{\min\left\{ \frac{1}{2}x - 1 \text{ subject to } -5 \leq x \leq 5 \right\}}_{=-\frac{7}{2}}$$

- Both minima are attained at $x = -5$ (black points in the figure on the next slide)

Example (2/2)



Property: multiplication by a negative constant

- For every LP problem and every $a \in \mathbb{R}_{<0}$, we have:

$$\min \left\{ a\, \boldsymbol{c}'\boldsymbol{x} \text{ subject to } \boldsymbol{x} \in F \right\}$$

$$= a \max \left\{ \boldsymbol{c}'\boldsymbol{x} \text{ subject to } \boldsymbol{x} \in F \right\}$$

- Multiplying by a negative constant turns the graph upside down: the points of minimum of $a\, \boldsymbol{c}'\boldsymbol{x}$ are the points of maximum of $\boldsymbol{c}'\boldsymbol{x}$

Example (1/2)

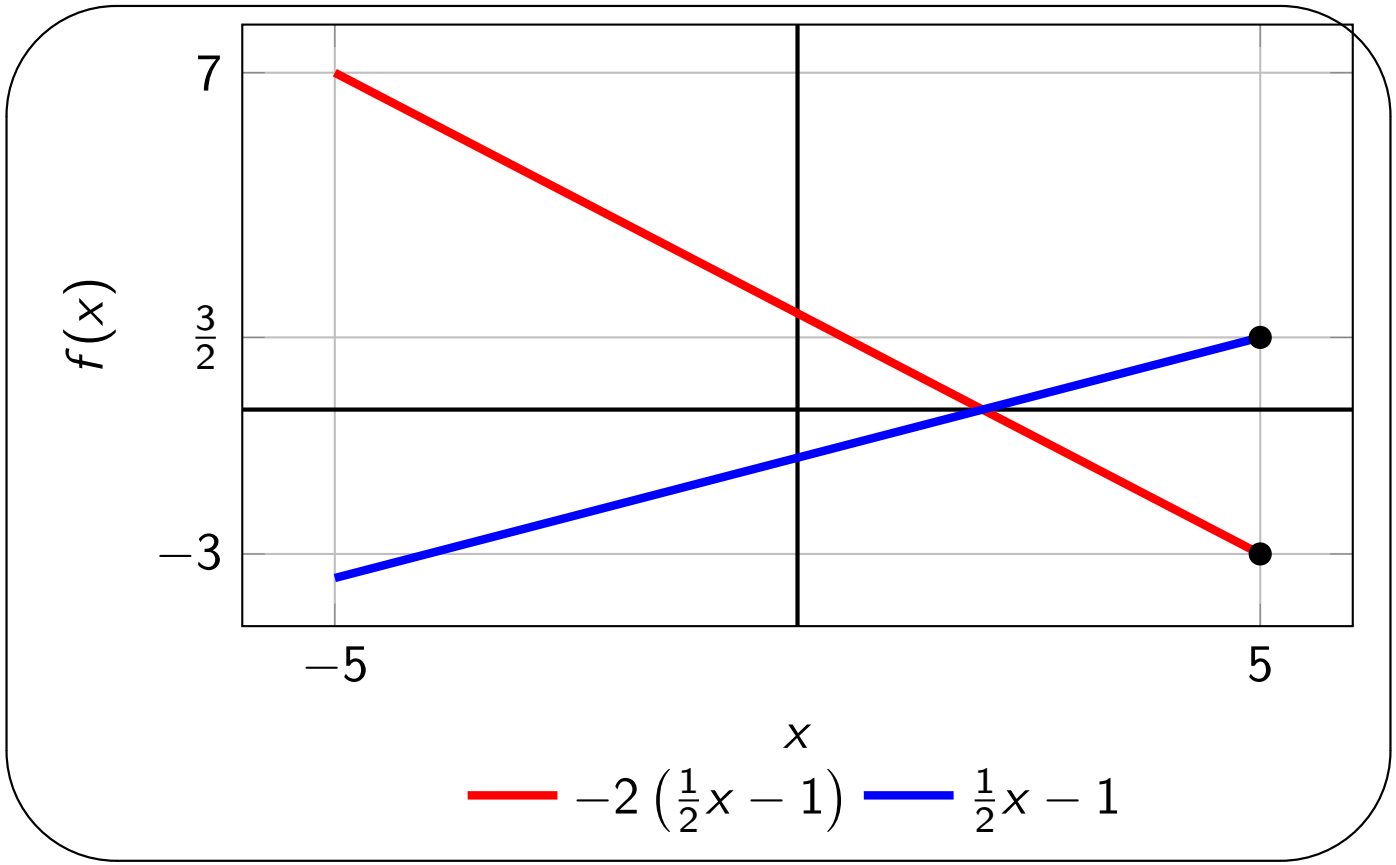
Property: multiplication by a negative constant.

$$\underbrace{\min\left\{ -2\left(\frac{1}{2}x - 1\right) \text{ subject to } -5 \leq x \leq 5 \right\}}_{=-3}$$

$$= -2 \underbrace{\max\left\{ \frac{1}{2}x - 1 \text{ subject to } -5 \leq x \leq 5 \right\}}_{=\frac{3}{2}}$$

- The minimum of $-2\left(\frac{1}{2}x - 1\right)$ and the maximum of $\frac{1}{2}x - 1$ are both attained at $x = 5$ (black points in the figure on the next slide)

Example (2/2)



Summary

An LP problem in one variable:

- A strictly increasing objective function over an interval: the minimum is at the left endpoint, the maximum at the right endpoint

Four properties of LP problems:

- $\min\{\, \boldsymbol{c}'\boldsymbol{x} \,\} = -\max\{\, -\boldsymbol{c}'\boldsymbol{x} \,\}$
- Adding a constant a adds a to the optimal value
- Multiplying by $a > 0$ multiplies the optimal value by a
- Multiplying by $a < 0$ turns a minimization into a maximization
- In the first three cases the optimal solutions do not change; in the last one, the points of minimum of $a\,\boldsymbol{c}'\boldsymbol{x}$ are the points of maximum of $\boldsymbol{c}'\boldsymbol{x}$

Summary

- An **LP problem** minimizes or maximizes a linear objective function $f(\boldsymbol{x}) = \boldsymbol{c}'\boldsymbol{x}$ over a polyhedron F, the feasible region, described by $\boldsymbol{A}$, $\boldsymbol{b}$ and the type of each row
- The goal is an **optimal solution** $\tilde{\boldsymbol{x}}(LP)$; its value is the **optimal value** $z(LP)$
- An optimal solution does not exist only if the problem is **unbounded** or **infeasible**
- Every **feasible solution** gives a bound on $z(LP)$: an upper bound (min) or a lower bound (max)
- Changing sign, adding a constant or multiplying by a constant changes the optimal value in a simple way, and lets us pass from min to max